6. MASS AND BALANCE

6.1 HOIST LOAD CG CALCULATIONS

6.1.1 Arm

The longitudinal distance (arm) from station 0 (RD) to a load suspended from the hoist:

Boom extended . . . . 3200 mm (126 in)
Boom retracted . . . . 2500 mm (98 in)

6.1.2 Hoist Loading Chart (see Fig. 1)

The hoist loading chart is provided as an aid in determining:

– CG limits during hoist operation

– Max allowable load for a given inflight CG and gross mass.

NOTE Hoist load envelopes for this chart are calculated using limit values with loads attached to a retracted boom (most forward CG condition).

EXAMPLE: **1**

Determine: **CG limits during hoist operation**

Known:

Helicopter inflight gross mass (without hoist load)	1700	kg
Helicopter inflight CG	3160	mm
Hoist load	200	kg

Solution:
1. Enter chart at known inflight gross mass (1700 kg).
2. Move right to intercept known inflight CG (3160 mm).
3. Move to the left to intercept the nearest limit line and read max allowable hoist load = 200 kg.

Since the helicopter inflight gross mass/CG point lays within the applicable hoist load CG envelope (200 kg), the helicopter CG will remain within the permissible range when hoisting this load.

EXAMPLE: **2**

Determine: **Max allowable load for a given inflight gross mass and CG**

Known:

Helicopter inflight gross mass (without hoist load)	1700	kg
Helicopter inflight CG	3120	mm

Solution:
1. Enter chart at known inflight gross mass (1700 kg).
2. Move right to intercept known inflight CG (3120 mm).
3. Move to the left to intercept the nearest limit line and read max allowable hoist load = 80 kg.